THE FOUR PHYSICAL CONSTRAINT COLUMNS OF PHYSICAL AI

The Unforgiving Laws of Physics, Mechanics, Thermodynamics, and Silicon Under Which Cognitive Agents Must Operate

COLUMN 1

Time & Freshness

Governing Physics:

Information Age: $\Delta t = t_{act} - t_{sens}$
World Deadline: τ_{world}

Uncertainty: $\sigma(t) = \sigma_0 + v\Delta t$

Critical Systems Metrics:

- Freshness Wall Δt_{wall}
- Tail Latency $(P_{99}, P_{99.9})$
- Latest-Frame Sampling
- Action Chunking (H steps)

Primary Failure Mode:

Stale-state collapse; open-loop drift during latency spikes.

COLUMN 2

Inertia & Stopping

Governing Physics:

Momentum: $\mathbf{p} = m\mathbf{v}$
Kinetic Energy: $E_k = \frac{1}{2}mv^2$

Braking Decel: $a_{max} \leq \mu g$

Critical Systems Metrics:

- Stopping Envelope d_{stop}
- Linear Reaction: $v \cdot \Delta t$
- Quadratic Brake: $\frac{v^2}{2a_{max}}$
- Clearance Barrier: d_{clear}

Primary Failure Mode:

Kinetic overrun; collision with obstacles during deceleration.

COLUMN 3

Mechanical & Thermal

Governing Physics:

Torque Rate: $\dot{\tau} = d\tau/dt$
Jerk Continuity: $\dddot{\mathbf{q}} = d^3\mathbf{q}/dt^3$

Joule Heating: $C_{th}\dot{T} = I^2R - \frac{\Delta T}{R_{th}}$

Critical Systems Metrics:

- Gearbox Shear Stress
- C^2 Jerk-Continuous Splines
- Continuous Current I_{cont}
- I^2t Thermal Winding Clamp

Primary Failure Mode:

Gearbox tooth stripping; stator winding thermal burnout.

COLUMN 4

Silicon Determinism

Governing Physics:

Crossbar Saturation: UMA DRAM
Ingestion Tax: DMA vs NPU

SRAM Mailbox: Lock-Free Ring

Critical Systems Metrics:

- Pre-Inference Bus Tax
- Heterogeneous Coherency
- Zero-Heap Real-Time Rule
- Hardware Watchdog Leases

Primary Failure Mode:

Memory bus starvation; non-deterministic heap pause.

THE CO-DESIGN INVARIANT: Every cognitive action proposed by high-level software must be verified against all four columns before motor gate drivers are energized.